Comprehensive Analysis of Input Order Invariant Approximate 4-2 Compressors for Binary Multipliers

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***Abstract*—Approximate arithmetic circuits sacrifice computing accuracy in exchange for improvements in power, area, and speed. Many approximate binary multipliers that use 4-2 compressors have been proposed but most of the proposals have error performances that depend on the order in which the inputs are connected to the compressors. This complicates the design and prevents a fair comparison among different approximate multipliers. The paper proposes the input order invariant approximate 4-2 compressors, whose behavior remains consistent regardless of the order of input signals. We derive the complete set of such compressors and utilize them to synthesize 8-bit multipliers. Our analysis reveals that only a limited subset of input order invariant approximate 4-2 compressors offers an optimal balance between error and power savings. Furthermore, we demonstrate that this optimal set of 4-2 compressors can be strategically distributed within the columns of the multiplier to further enhance the trade-off between error and power efficiency. The proposed circuits, once implemented in a 14 nm FinFET standard cell technology, favorably compare against the state of the art.**

***Index Terms*—approximate computing, approximate compressors, unsigned binary multiplier, low power digital circuits**

I. Introduction

Approximate computing is an emerging field that focuses on arithmetic circuits designed to provide efficiency gains in terms of power, even if their results are not always correct. These circuits are primarily intended for error-resilient applications, including machine learning, data mining, and image processing [1]–[5].

Hardware-level approximation has been investigated for many arithmetic blocks with a particular attention to the binary multiplier that constitutes one of the most energy-hungry digital blocks [6]. Several approaches for the design of approximate multipliers have been proposed in literature [7], [8]. Approximate recursive multipliers use small approximate modules that are assembled to obtain n×n multipliers [9]–[13]. In truncated multipliers, [14], [15] some of the partial products are not formed and the resulting truncation error is mitigated with the help of suitable correction functions. Logarithmic multipliers [16], [17] approximate the multiplication via log-summation-antilog stages.

Utilizing approximate compressors is a widely adopted approach in the design of approximate multipliers. A compressor counts the number of ones in its inputs and encodes the result in its outputs. In a multiplier, a tree of compressors is employed to reduce the process of summing the partial products into a two-operand addition. The full adder is the most basic compressor, that converts its three inputs into a count value, encoded in two outputs (carry and sum). Higher-order compressors are also frequently used in multipliers, with the most common choice being the 4-2 compressor [18]–[20].

Many papers have recently proposed several versions of approximate 4-2 compressors, that can be efficiently employed to build low-power approximate multipliers [21]–[35]. However, most of them exhibit varying error performance when used in approximate multipliers, depending on the order their inputs are connected to the set of bits to be compressed [32].

The inconstant error behavior of such approximate compressors is an additional complication for the designer hoping to select the most suitable circuit for a given application. Aiming to alleviate this burden, this paper investigates input order invariant approximate 4-2 compressors, whose behavior remains consistent regardless of the order of input signals.

Utilizing these compressors to construct approximate multipliers leads to circuits with errors independent of the wiring between compressors, ensuring consistent behavior that can be easily replicated by every designer.

The results of the paper can be summarized as follows:

- We describe the set of input order invariant approximate 4-2 compressors, composed by 1024 different circuits.
- For each approximate compressor of the set, we construct an 8-bit approximate multiplier and calculate its error and electrical performance considering a 14 nm FinFET standard-cell technology.
- From the analysis of the characteristics of the 1024 8-bit approximate multipliers, we found that only a small set of

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input order invariant approximate 4-2 compressors offers an optimal balance between error and power savings. It should be noted that power dissipation exhibits a strong correlation with the silicon footprint, particularly in such compact designs. Hence area occupation is not considered as a performance metric in this paper.

- We demonstrate that this optimal set of 4-2 compressors can be strategically distributed within the columns of the multiplier, to further enhance the trade-off between error and power efficiency.
- We compare the resulting 8-bit multipliers with the state of the art showing that the proposed circuits equal or improve the performances of the previously proposed 4-2 compressors with the additional benefit of providing reproducible performance.

II. Input Order Invariant Approximate 4-2 Compressors

A. Approximate 4-2 compressors

The exact 4-2 compressor, [20], should be better referred to as a 5-3 counter since it has four input bits ($x1$,$x2$,$x3$,$x4$) and a carry in bit (cin), all of them with weight one, and reports their sum on a sum bit (s) of weight one, a carry bit (c) and a further carry out ($cout$) bit, that weigh two. The $cout$ does not depend on cin thus avoiding the carry propagation when two or more compressors are used in cascade.

The approximate 4-2 compressor disregards the cin and $cout$ bits, leaving only s and c as outputs. This circuit inherently contains an unavoidable error because the maximum sum of the input bits is four, while the maximum possible result that can be encoded in the output is three.

Additional errors can be intentionally introduced into the truth-table of an approximate 4-2 compressor, aiming to simplify its hardware implementation. Some of the approximate 4-2 compressors found in the literature are reported in the first five columns of Tab. I. The entries that provide inaccurate results are highlighted in bold.

TABLE I

Truth Tables for some Approximate 4-2 Compressors.

Input	Output					
	[24]	[28]	[29]	[32]	[33]	01133[a]
$x_1x_2x_3x_4$	cs	cs	cs	cs	cs	cs
0000	00	00	**01**	00	00	00
0001	01	01	01	01	01	01
0010	01	01	01	01	01	01
0011	10	10	**01**	10	10	**01**
0100	01	01	01	01	01	01
0101	**01**	10	**11**	10	10	**01**
0110	**01**	10	**01**	10	10	**01**
0111	11	11	11	**10**	11	11
1000	01	01	01	01	01	01
1001	**01**	10	**11**	10	10	**01**
1010	**01**	10	**01**	10	10	**01**
1011	11	11	11	11	11	11
1100	10	**01**	**11**	10	10	**01**
1101	11	**10**	11	11	11	11
1110	11	**10**	11	11	11	11
1111	**11**	**11**	**11**	**11**	**11**	**11**

[a]The meaning of the ID number is explained in Section II-B.

Most of the approximate 4-2 compressors proposed in the literature are not input order invariant, meaning that the error of the circuit depends on the order in which the input signals are connected to the ports of the compressor. Let us consider, as an example, the approximate 4-2 compressor of [24] (first column of Tab. I). This circuit produces the correct output c=1, s=0 when the input $x_1 \ldots x_4$ is 0011. However, it gives an erroneous result c=0, s=1 when the input $x_1 \ldots x_4$ is 0101, indicating that inputs x_2 and x_3 are treated differently. As an additional example, in the approximate compressor [28] (second column of Tab. I), the input $x_1 \ldots x_4$ 0111 gives the output c=1, s=1, while the input $x_1 \ldots x_4$ 1101 (still having three bits high) gives a different output c=1, s=0.

B. Input order invariance

Let us consider the approximate 4-2 compressor from [33] whose truth table is also reported in Tab. I. This circuit is input order invariant: it gives the same output c=0, s=1 for all the input combinations with only one input high (no matter which input is high), the same output c=1, s=0 for all the input combinations with two inputs high (no matter which couple of inputs is high) and so on. The same can be said for the circuit labeled as 01133 and shown in the last column of Tab. I, that has a larger number of erroneous entries but still holds the property of being input order invariant.

In general, the distinguishing feature of input order invariant 4-2 compressors is the following:

- the output of the compressor is the same for all the four truth table entries that have only one bit high. $\{x_1,x_2,x_3,x_4\}$=$\{1,0,0,0\}$,$\cdots$, $\{0,1,0,0\}$, etc.
- the output of the compressor is the same for all the six truth table entries that have two bits high. $\{x_1,x_2,x_3,x_4\}$=$\{1,1,0,0\}$,$\cdots$, $\{0,1,1,0\}$, etc.
- The output of the compressor is the same for all the four truth table entries that have three bits set high: $\{x_1,x_2,x_3,x_4\}$=$\{1,1,1,0\}$, $\cdots$, $\{0,1,1,1\}$, etc.

In addition, the circuit is also defined by the output provided when all the input bits are 0 or 1. For each case listed above, the output can be $\{0,0\}$, $\{0,1\}$, $\{1,0\}$, $\{1,1\}$ corresponding to the decimal numbers 0, 1, 2, and 3. In summary, for an input order invariant approximate 4-2 compressor, the 16 input combinations of x_1,x_2,x_3,x_4 can be reduced to five groups, each of which produces c and s outputs that (seen as decimal numbers) can be 0, 1, 2, or 3. This gives a total of $4^5 = 1024$ different input order invariant approximate 4-2 compressors.

Each input order invariant 4-2 compressor can be uniquely identified by a five-digits ID. Each digit in the ID signifies the obtained output for each of the five cases indicated above (the leftmost digit denotes the case where all inputs are set to 0, moving on to the case with one input set high, and so on). For instance, the circuit with the truth table provided in the rightmost column of Tab. I has ID=01133. Similarly, one can easily verify that the truth table in the second-to-last column of Tab. I, [33], is an input order invariant circuit with ID=01233.

TABLE II

MRED VARIATION FOR 8×8 MULTIPLIERS, VARYING THE ORDER IN WHICH SIGNALS ARE CONNECTED TO THE PORTS OF THE APPROXIMATE COMPRESSORS.

4-2 compressor	MRED minimum	MRED maximum
[24]	1.3×10^{-2}	1.8×10^{-2}
[28]	3.4×10^{-3}	9.4×10^{-3}
[29]	9.0×10^{-2}	0.3×10^{-1}
[32]	7.7×10^{-4}	2.1×10^{-3}
01232, [26]	4.7×10^{-4}	
01233, [33]	2.4×10^{-4}	

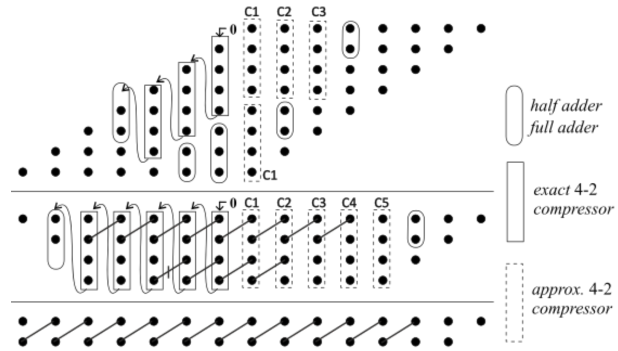

Fig. 1. Reduction scheme for the 8×8 unsigned multiplier. The approximate 4-2 compressors are on columns 7, 6, 5, 4, and 3. C1, C2, etc. indicate the type of employed approximate 4-2 compressor, as explained in Section IV.

To highlight the significance of employing input order invariant approximate compressors, we implemented several 8×8 multipliers by using approximate 4-2 compressors in the right-half of the partial product matrix, as described in Sec. III. We explored various versions of each multiplier, where each version retains the same approximate compressor but different order in which the different signals are connected to the ports of the approximate compressors.

For an n-bit unsigned multiplier, the relative error distance (RED) is defined as:

$$\text{RED} = \frac{|M_i - Mapp_i|}{M_i} \quad (1)$$

for $M_i \neq 0$, where M_i is the exact value of the i-th multiplication and $Mapp_i$ is the value produced by the approximate multiplier. MRED is defined as the average value of RED.

Table II reports the minimum and maximum MRED values obtained for the various versions of the multipliers. The multipliers that do not employ input order invariant approximate compressors exhibit significant variations in MRED, depending on the order of the wire connections to the ports of the approximate compressors. For instance, different implementations using the approximate compressor proposed in [29] show a threefold variation in MRED. On the contrary, the use of input order invariant approximate compressors consistently yields the same MRED value, ensuring repeatable results.

III. Finding the Best Input Order Invariant Approximate 4-2 Compressors

As explained in the previous section, there is a total of 1024 different input order invariant approximate 4-2 compressors. This number, while large, allows a brute force analysis for the identification of the best performing circuits.

To obtain realistic results, we used as a testbench an approximate 8×8 multiplier, whose structure is shown in Fig. 1. Similarly to previous papers [32] the approximate 4-2 compressors are employed in the 8 right-most columns of the multiplier, while exact 4-2 compressors are used in the remaining columns.

We have described in HDL and simulated 1024 multipliers (each one using a different input order invariant approximate 4-2 compressor) to obtain the MRED, as error metric. Then, we have synthesized each multiplier, targeting a 14 nm FinFET standard cell technology (0.8 V supply voltage, regular V_T).

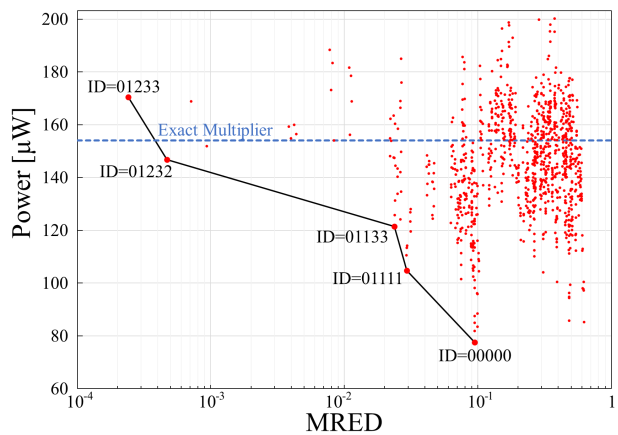

Fig. 2. Power dissipation vs MRED for 8×8 approximate multipliers. Each dot in this Figure represents a multiplier using a different input order invariant approximate 4-2 compressor. Only the few compressors on the Pareto front give the best power-error trade off.

The synthesis is conducted with the Cadence Genus software, with a timing constraint set to the value of 1 ns. The power dissipation is computed by simulating the synthesized netlist with the input toggle frequency of 1 GHz, outputs loaded with a NAND2X cell, and applying 2^{16} random input combinations to obtain the switching activity.

The syntheses were performed with a workstation equipped with a 72-cores Xeon CPU E5 v3 2.3GHz, with 132GB RAM. Each synthesis takes about 20s, so that the total CPU time to perform all the syntheses is in the order of 6 hours.

The obtained results are plotted in Fig. 2, where the power dissipation of each multiplier is reported as a function of the MRED, an error metric commonly used in the literature [12], [13], [16], [24], [32], that is a good overall indicator of circuit performance in error resilient applications. Each dot in this figure represents a multiplier using a different input order invariant approximate 4-2 compressor. As a reference, the power of the multiplier using exact 4-2 compressors is about 154 μW at 1 GHz toggling frequency. The multipliers with the best trade-off between power consumption and precision are those closest to the lower-left corner of the figure. As it can be observed, most of the investigated compressors are sub-

TABLE III
SELECTED INPUT ORDER INVARIANT 4-2 COMPRESSORS.

Compressor		8×8 multiplier	
ID	Standard cell implementation (gates)	MRED	Power [μW]
01232	5 gates: XOR2, XOR3, NAND2, inv. Carry, OAI	4.68×10^{-4}	146.75
01133	6 gates: 3 NAND2, 2 NOR2, OAI	2.36×10^{-2}	121.39
01111	1 gate: OR4	2.93×10^{-2}	104.78

optimal since they either dissipate more than the exact circuit or have error larger than the circuit with ID=00000 which corresponds to the truncation of the partial products. The circuits on the Pareto front are only those with the following IDs: 01233 (already proposed in [33]), 01232 (already proposed in [26]), 01133, 01111, and 00000. Within this subset, we excluded the compressor 01233, as it exhibited higher power dissipation than the exact circuit, and the trivial design 00000. In summary, our analysis allows identifying three optimal input order invariant approximate 4-2 compressor, as reported in Tab. III where the performances and a summary of the standard cell implementation are reported.

IV. Optimal Approximate 8×8 multiplier

Instead of using the same approximate 4-2 compressor in all the columns of the multiplier, we can further enhance the trade-off between error and power efficiency, as well as better populate the Pareto front by strategically distributing the approximate compressors within the columns of the multiplier. In particular, it makes sense to use the most accurate compressors in the left-most columns of the multiplier and the less accurate ones in the right-most columns.

We implemented all the possible combinations, distributing the three approximate compressors of Tab. III in the columns C1 through C5 shown in Fig. 1. The results giving the best trade-off between power reduction and MRED are reported in Fig. 3. The multipliers indicated in this figure with the letters A through I use the compressors configurations reported in Tab. IV. The most accurate multiplier indicated with the 'ID=01232' label uses the 01232 approximate compressor in every column while the least power hungry indicated with the 'ID=01111' only uses the 01111 approximate compressor.

Fig. 3 includes results obtained by using previously proposed approximate 4-2 compressors. Since in this case the error is influenced by the order in which the input signals are connected to the ports of the compressors, these multipliers are represented by horizontal segments, with their abscissa ranging from the minimum to the maximum MRED values we obtained. In addition we implemented the two non input order invariant designs proposed in [35]. They are located outside the boundaries of Fig. 3 with MRED and power values of 0.54, 33.5 μW for *proposed-mul1* and 0.18, 45.6 μW for *proposed-mul2*. As expected, the error values differ from those published in [35].

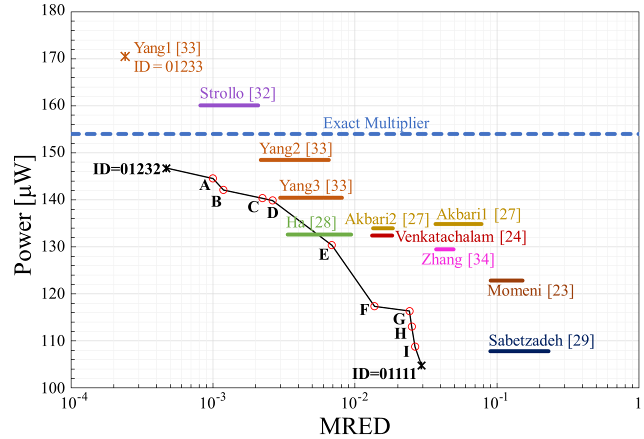

Fig. 3. Power dissipation vs MRED using different arrangements of approximate 4-2 compressors, (letters A through I). The distribution of compressors among the columns of the multiplier is in Tab. IV. Horizontal segments refer to previously published approximate multipliers, having error depending on the order in which the input signals are connected to the ports of the compressors.

TABLE IV
STRUCTURE OF THE PROPOSED APPROXIMATE 8×8 MULTIPLIERS USING INPUT ORDER INVARIANT 4-2 COMPRESSORS.

	C1	C2	C3	C4	C5	MRED $\times10^{-3}$	Power [μW]
A	01232	01232	01232	01232	01133	0.99	144.56
B	01232	01232	01232	01232	01111	1.18	142.10
C	01232	01232	01232	01133	01111	2.22	140.36
D	01232	01232	01232	01111	01111	2.63	139.86
E	01232	01232	01111	01111	01111	6.86	130.35
F	01232	01111	01111	01111	01111	13.7	117.35
G	01133	01133	01133	01111	01111	24.2	116.36
H	01133	01133	01111	01111	01111	25.1	113.02
I	01133	01111	01111	01111	01111	26.5	108.77

The multipliers with the proposed configurations obtain in most cases better error-power trade-off compared to previous proposals, with the additional advantage of having an error independent from the wiring of the approximate compressors, ensuring repeatable results.

V. Conclusion

We have proposed the input order invariant approximate 4-2 compressors, whose behavior remains consistent regardless of the order of input signals. We have also performed an exhaustive analysis of the entire set of these circuits.

Our analysis reveals that a small subset of these compressors offers an optimal balance between error and power savings. Furthermore, we demonstrate that this optimal set of 4-2 compressors can be strategically distributed within the columns of the multiplier to further enhance the trade-off between error and power efficiency.

The approximate multipliers using the configurations in Tab. IV allow the designer to choose the best suited circuit for the desired application, with a power reduction and a MRED from about 6% and 10^{-3}, to about 30% and 27×10^{-3}, respectively, without worrying about error dependencies.

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